MILITARY SPECIFICATION

INTEGRATED CIRCUIT, DIGITAL, COMOS, TIME BASE AND SPECIAL PURPOSE FUNCTION GENERATOR, MONOLITHIC, SILICON

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for a single, monolithic, complementary symmetry metal-oxide-semi-conductor (COMOS) microcircuit to be used as a low power time base and special purpose function generator in a family of scatterable mine assemblies. It is intended for high shock environments.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or, request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38510	-	Microcicuits, General Specification for
MIL-A-48078	-	Ammunition Standard Quality Assurance Provisions, General Specification for
MIL-M-55555	-	Microcircuits, Preparation for Delivery of
MIL-S-45743	-	Soldering, Manual Type

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-46844	Solder Bath Soldering of Printed Circuit Assemblies
MIL-P-48189	Housing, Timing & Fuzing (ADAM)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-1313	Microelectronic Terms and Definitions
MIL-STD-1331	Parameters to be Controlled for the Specification of Microcircuits

DRAWINGS

us ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

INTEGRATED CIRCUIT, CONTROL AND TIMING BASE CMOS, 9298507

Tests Vehicle, Ballistic Rail Gun 9298426
Housing, Timing and Fuzing 9298450

3. REQUIREMNETS

3.1 General. The manufacturer of microcircuits in compliance with th is specification shall have production and test facilities and a quality and reliability assurance program adequate to assure successful compliance with the provisions of this specification. Only microcircuits which are inspected for and meet all the requirements specified herein shall be marked and delivered.

3.2 Terms, definitions and symbols. For the purpose of this specification, the terms, definitions and symbols in MIL-STD-893, MIL-STD-1313, MIL STD-1331, MIL-M-38510 and those contained herein shall apply.

3.3 Material. Materials shall be in accordance with the applicable drawings and specifications

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3.4 Physical dimensions. The integrated circuit shall compIy with all the requiremen ts specified in Drawing No. 9298507, all associated drawings, and with the requirements specified in MIL-STD-883, Method 2016.

3.5 Soldrability. The I.C. leads shall comply with the solderability requirements of MIL-STD-883, Method 2003.

3.6 Resistance to solvents. The I.C. marking shall not become illegible when subjected to solvents and comply with the resistance to solvents requirements of MIL-STD-883, Method 2015.

3.7 Bond strength. Wires from the circuit die to the component wire bonds shall be capable of meeting the bond strength requirements of MIL-STD-883, Method 2011., Condition D. Pull all bonds on five (5) devices. There small be no failures less than one (1) gram.

3.8 Die bond shear strength. The integrated circuit shall comply with the die band shear strength requirements of MIL-STD-883, Method 2019.

3.9 Pre-cap visual. The integrated circuit (IC) shall comply with the pre-cap requirements of MIL-STD-883, Method 2010.2 - test condition B except delete paragraphs 3.2.1.7, 3.2.2, 3.2.6, 3.2.7, 3.2.8 and 3.2.9.

3.10 Lead fatigue. The IC package shall comply with the lead fatigue requirements of MIL-STD-883, Method 2004, Test condition B2. (Piece part level)

3.11 Seal fine and gross. The IC shall meet the requirements of MIL-STD-833 method 1014 conditions A or B and C.

3.12 Conditioning screen. The integrated circuit shall be conditioned through the screening procedures listed in paragraph 4.4.2.2 and in accordance with the applicable test methods of MIL-STD-883.

3.13 Dynamic burn-in test. The IC shall be burned-in at a temperature of one hundred and twenty five degrees centigrade plus or minus five degrees centigrade (125°C + 5°C) for a minimum time of forty eight (48) hours with the IC connected as shown in figure 18.

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3.14 Post burn-in electrical parameters. The IC shall meet the requirements of Table No. II II when it is tested at a temperature of twenty five degrees centigrade plus or minus five degrees centigrade (25°C + 5°C). Input waveforms of Fig 6 shall be applied and output waveforms of Fig 7 shall be observed. See paragraph 4.4.2.3.

3.14.1 Leakage current. The IC, when connected as shown in figures 9a and 9b, shall not exceed the requirements as required in Tables II, III & IV.

3.14.2 Oscillator function current. The IC, when connected as shown in figure 4,shall function as an oscillator as required in Tables 11, III & IV.

3.14.3 Output period. The output period shall not shift more than plus five percent or minus seven percent from the period measured at twenty five degrees (25°) centigrade and VDD=4.O volts when the temperature is lowered to minus forty degrees (-40°) centigrade and then raised to plus sixty degrees (+60°) centigrade and the supply voltage, VDD varies from 3.5 to 4.0 volts.

3.14.4 Oscillator output voltage, terminals 3 and 11. The output voltages on terminals 3 and 11 shall be pulses with the following characteristics as defined in MIL-STD-1331, Appendix A.

Amplitude	VDD-0.25 volts, minimum
Rise time	1.0 microseconds, maximum
Fall time	1.0 microseconds, maximum
Pulse Duration	1.0 milliseconds, minimum

The rise and fall times of the pulses on terminals 1, 3, 9, and 11 shall be smooth transitions with no irregularities between the 20% and 80% points of full amplitude. The waveforms shall be evaluated with an oscilloscope having a ten megohm probe. No other instrumentation shall be connected other than that shown in Fig. 4.

3.14.5 "0" output voltage: terminal 12 (Battery Load). The IC, when connected as shown in fig. 12, shall meet the requirements of Tables II, III & IV.

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3 .14.6 "1" output voltage: terminals 13 (Initiator) and 15 (Detonator). The IC, when connected as shown in fig. 10, shall meet the requirements of Tables II, III & IV.

3.14.7 "0" output voltage: terminals 13 (Initiator) and 15 (Detonator). The IC, when connected as shown in fig. 11, shall meet the requirements of Tables II, III & IV.

3.14.8 Input switching voltage, terminals 2 (F1) and 10 (F2). See fig. 14. A slowly increasing voltage is applied to terminal 2 or 10. The level on pin 1 or 9 shall be observed and when pin 1 or 9 goes from a low to a high level, the voltage level on terminal 2 or 10 shall be measured as the threshold voltage on pin 2 or 10.

3.14.9 Sequential operation. The IC, when connected as shown in fig. 5 and the waveforms shown in fig. 6 are applied to terminals 1, 2, 4, 5, 6, 7, 16, 9, 10 and 14, shall operate as described by the output waveforms in fig. 7.

3.14.10 Sequential operation waveform characteristics. All pulse magnitudes (figures 6 and 7) are equal to VDD except on pin 4 as shown on figure 6 and explained in 3.2.7.11. All pulse counts are referenced to the zero start time. One pulse count is equal to one pulse of the oscillator (terminals 1 and 2).

3.14.11 Input terminal number four (4) (LVD). The input voltage levels applied to the input terminal number four (4) during the sequence test must be the maximum and minimum LVD voltages specified in Table II. See fig. 13.

3.14.12 Input resistance. The input resistance on terminals 5,6, and 7 shall be measured by connecting the device as shown in fig. 15 and meet the requirement in Table II.

3.14.13 "1" output voltage: terminal 1, 3 (F1), 9 and 11 (F2). The IC, when connected as shown in fig. 16, shall meet the requirements of Table II.

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3.15 Temperature. The IC shall be capable of operating over the temperature l imits of minus fifty five degrees centigrade (-55° C) to plus eighty degrees centigrade (+80°C) first article tests, and minus forty degrees centigrade (-40°C) to plus sixty degrees centigrade (+60°C), lot acceptance tests. See Tables I, III & IV also paragraph 4.4.2.4 and 4.4.2.5.

3.16 Country of origin. The IC shall be manufactured and sold by a qualified producer in the United States or in another country which is a party to one of the existing international agreements ratified by the United States, such as the NATO STANAG-4093, Mutual Acceptance of Qualification Approvals for Electronic Parts, and the United States/Canada/ Australia Agreement for Qualification of Products for Non-resident Manufacturers.

3.17 Group C environmental requirements. The IC shall comply with the requirements of paragraph 4.4.2.6 and to the applicable test method of MIL-STD-863 If more than two (2) relevant failures result from the first or combined sample, the I.C. supplier will perform failure analysis to determine cause and institute corrective action. A copy of the failure analysis report is to be furnished to the technical agency listed in paragraph 6.5.

3.18 Mechanical shock, ballistic rail gun. The IC when mounted in a electrically functionable ADAM wedge and loaded into a 155MM test vehicle projectile, shall be capable of surviving the mechanical shock associated with a zone 7X firing or equivalent from a 155MM howitzer. In addition, the IC shall be capable of surviving cargo ejection with subsequent ground impact in a random orientation of an uncontrolled surface which may include rocks or other hard matter. The wedge shall be electrically tested as per MIL-P-48189(AR).

3.19 Data requirements. Data shall be recorded and retained for all test required by this specification as specified in 4.5.17.11.

3.20 Process controls. The vendor will manufacture IC devices in an approved MIL-M-38510 facility and will adhere to the appropriate MIL-M-38510 documents. In the event of any non-conformance, the following should occur.

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e. Contractor to investigate, perform failure analysis and take corrective action.

b. Assessment of Material processed to data for conformance.

c. Contractor to supply documentation.

3.21 Workmanship. The integrated Circuit shall be fabricated and finished in a thorough, workmanlike manner. It shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to the Integrated Circuit nor shall the parts be contaminated by the cleaning agents. All markings and stampings shall be free of harmful cleaning and soldering agents that may be injurious to the IC. All markings and stampings, shall be displayed and visibly defined. The IC shall be manufactured and/or processes in such a manner as to be uniform in quality and free from defects that will affect life, configuration and serviceability.

3.22 Qualification. The integrated circuit shall be subjected to another first article if:

a. The contractor's manufacturing techniques or process controls which affect the form, fit, function or internal geometry of the IC are revised by either the contractor or one of his sub-contractors supplying parts or assemblies.

b. The contractor relocates his assembly facilities.

c. The contractor uses a new sub-contractor for material, parts or assembly.

d. A change in material used for integrated circuit process or assembly.

Qualification requirements shall be determined by the Government for each change, (See Table I, requirements).

3.23 First article inspection. This specification contains provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be specified in the contract, see Table I.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection.

b. Quality conformance inspections.

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall be consistent with the First Article Classification of Defects, Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and First Article Inspection Classification of Defects, Table I and drawings as applicable.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the classification of defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Integrated Circuit, Digital, CMOS, Time Base and Special Purpose Function Generator		SHEET 1 OF 3		DRAWING NUMBER 9298507
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for defects				4.4.2.1, 4.4.2.2, 4.4.2.3 4.4.2.4, 4.4.2.5, 4.4.2.6
	Resistance to Solvents	6		3.6	4.5.8
	Solderability	6		3.5	4.5.7
	Physical Dimensions	6		3.4	4.5.1
	Pre-Cap Visual (DE-LID)	36(a)		3.9	4.5.6
	Bond Strength (DE-LID)	6		3.7	4.5.2
	Die Bond Shear Strenght (DE-LID)	6		3.8	4.5.4
	Lead Fatigue	6		3.10	4.5.3
	Country of Origin		100%	3.16	Certification
	* Conditioning Screen	36(a)		3.12	4.5.5
	* Dynamic Burn-In	36(a)		3.13	4.5.9
	Electrical Parameters, Room Ambient, 25C	36		3.14	4.5.10
	Electrical Parameters, Temperature	36		3.15	4.5.17
	Mechanical Shock, Ballistic Rail Gun	36(b)		3.18	4.5.28
	* Integrated circuits submitted for First Article shall have been subjected to the conditioning screen and dynamic burn-in tests prior to submission and verified by the Government.				

NOTES:
(a) Due to the possibility that there might be some I.C. fall-out during these tests the contractor should plan on an initial quantity greater than 36.
(b) 36 IC's are required for ballistic rail gun testing.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: High Temperature Electrical Parameters (+80°C)		SHEET 2 OF 3		DRAWING NUMBER: 9298507
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Leakage current	36-0-1		3.15	4.5.17, Note 3
102	Oscillator function current	36-0-1		3.15	4.5.17
103	Oscillator function output period	36-0-1		3.15	4.5.17
104	"0" output voltage, terminals 13 & 15	36-0-1		3.15	4.5.17, Note 1
105	"0" output voltage, terminal 12	36-0-1		3.15	4.5.17, Note 2
106	Sequential operation	36-0-1		3.15	4.5.17, Note 3
107	LVD function voltage	36-0-1		3.15	4.5.17
Minor					
201	Evidence of poor workmanship		1.0%	3.21	Visual

NOTES:
(1) Measure after applying 4.0V (VDD) and then lowering VDD to 3.0V and before 176 pulses have been applied.
(2) VDD = 3.0V or 0.5V above LVD, whichever is greater.
(3) Due to possibility of IC failure, an initial quantity greater than 36 should be planned.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 9298507
	Lot Temperature Electrical Parameters (-55°C)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Oscillator function current	36-0-1		3.15	4.5.17, Note 3
102	Oscillator function output period	36-0-1		3.15	4.5.17
103	"1" output voltage terminals 13 & 15	36-0-1		3.15	4.5.17, Note 1
104	Sequential operation	36-0-1		3.15	4.5.17, Note 2
105	LVD function voltage	36-0-1		3.15	4.5.17
Minor					
201	Evidence of poor workmanship		1.0%	3.21	Visual

NOTES:
(1) Measure after applying 4.0V (VDD) and then lowering VDD to 3.0V and before 176 pulses have been applied.
(2) VDD = 3.0V or 0.5V above LVD, whichever is greater.
(3) Due to possibility of IC failure, an initial quantity greater than 36 should be planned.

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QUALITY CONFORMANCE INSPECTIONS

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Integrated Circuit, Digital, CMOS, Time Base and Special Purpose Function Generator (physical properties)				9298507
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Physical Dimensions	3-0-1		3.4	4.5.1
102	Solderability, 3 Devices, All 16 Leads	48-0-1		3.5	4.5.7
103	Bond Strength, 5 Pieces, All 16 Bondwires 1/	80-0-1		3.7	4.5.2
104	Die Bond Shear Strength, 5 Pieces 2/	5-0-1		3.8	4.5.4
105	Lead Fatigue, 5 Devices, All 16 Leads (Frame only) 3/	80-0-1		3.10	4.5.3
106	Resistance to Solvents	3-0-1		3.6	4.5.8
Minor					
201	Evidence of poor workmanship		1.0%	3.21	Visual

NOTES:
1. There shall be no failure less than 1 gram.
2. Check at piece part level, certification.
3. May be performed at pre-cap.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Conditioning Screen Tests				9298507
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pre-Cap Visual		100%	3.9	MIL-STD-883, Method 2010, Test Cond. B, except delete Para. 3.2.1.7, 3.2.2, 3.2.6 3.2.7, 3.2.8 and 3.2.9
102	Stabilization Bake		100%	3.12	MIL-STD-883, Method 1008, Test Cond. C
103	Temperature Cycling		100%	3.12	MIL-STD-883, Method 1010, Test Cond. C
104	Thermal Shock		100%	3.12	MIL-STD-883, Method 1011, Test Cond. C
105	Constant Acceleration		100%	3.12	MIL-STD-883, Method 2001, Test Cond. E
106	Seal, fine leak		100%	3.12	MIL-STD-883, Method 1014, Test Cond. A or B
107	Seal, gross leak		100%	3.12	MIL-STD-883, Method 1014, Test Cond. C
108	Dynamic Burn-in		100%	3.13	4.5.9
109	Electrical Parameters, D.C., 25°C		100%	3.14	4.5.10 except para. 4.5.10.2 4.5.10.3 and 4.5.10.4
Minor					
201	Evidence of poor workmanship		1.0%	3.21	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Electrical Parameters 25°C ± 5°C		9298507
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major:					
101	Leakage current	77-1-2		3.2.14.1	4.5.10.1
102	Oscillator function current (AC Test)	52-2-3		3.2.14.2	4.5.10.2
103	Oscillator function output period (AC Test)	52-2-3		3.2.14.3	4.5.10.3
104	Oscillator output voltage, terminals 3 & 11	52-2-3		3.2.14.4	4.5.10.4
105	"0" output voltage, terminal 12	77-1-2		3.2.14.5	4.5.10.5, Note 2
106	"1" output voltage, terminals 13 & 15	77-1-2		3.2.14.6	4.5.11.1
107	"0" output voltage, terminals 13 & 15	77-1-2		3.2.14.7	4.5.11.2, Note 1
108	Sequential operation	77-1-2		3.2.14.9	4.5.13, Note 2
109	LVD function voltage	77-1-2		3.2.14.11	4.5.14
110	Input switching voltage, terminals 2 & 10	77-1-2		3.2.14.8	4.5.12
111	Input resistance	77-1-2		3.2.14.12	4.5.15
112	"1" output voltage, terminals 1, 3, 9 & 11	77-1-2		3.2.14.13	4.5.16.1
113	"0" output voltage, terminals 1, 3, 9 & 11	77-1-2		3.2.14.14	4.5.16.2
Minor:					
201	Evidence of poor workmanship		1.0%	3.21	Visual

NOTES: (1) Measure after applying 4.0V (VDD) and then lowering VDD to 3.0V and before 176 pulses have been applied.
(2) VDD = 3.0V or 0.5V above LVD, whichever is greater.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	High Temperature Electrical Parameters (+60°C) Lot Acceptance Test Requirements	1 OF 1	9298507 NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Leakage current	52-2-3		3.2.14.1	4.5.10.1
102	Oscillator function current (AC Test)	52-2-3		3.2.14.2	4.5.10.2
103	Oscillator function output period (AC Test)	52-2-3		3.2.14.3	4.5.10.3
104	"0" output voltage, terminals 13 & 15	52-2-3		3.2.14.7	4.5.11.2, Note 1
105	"0" output voltage, terminal 12	52-2-3		3.2.14.5	4.5.10.5, Note 2
106	Sequential operation	52-2-3		3.2.14.9	4.5.13, Note 2
107	LVD function voltage	52-2-3		3.2.14.11	4.5.14
Minor					
201	Evidence of poor workmanship		1.0%	3.2.1	Visual

NOTES:
(1) Measure after applying 4.0 V (VDD) and then lowering VDD to 3.0 V and before 176 pulses have been applied.
(2) VDD = 3.0 V or 0.5 V above LVD, whichever is greater.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Low Temperature Electrical Parameters (-40°C) Lot Acceptance Test Requirements				9298507
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Oscillator function current (AC Test)	52-2-3		3.2.14.2	4.5.10.2
102	Oscillator function output period (AC Test)	52-2-3		3.2.14.3	4.5.10.3
103	"1" output voltage terminals 13 & 15	52-2-3		3.2.14.6	4.5.11.1, Note 1
104	Sequential operation	52-2-3		3.2.14.9	4.5.13, Note 2
105	LVD function voltage	52-2-3		3.2.14.11	4.5.14
Minor					
201	Evidence of poor workmanship		1.0%	3.21	Visual

NOTES:
(1) Measure after applying 4.0V (VDD) and then lowering VDD to 3.0V and before 176 pulses have been applied.
(2) VDD = 3.0V or 0.5V above LVD, whichever is greater.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Group C, Environmental - 3 Month/90 Day Interval				9298507
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Thermal Shock	52-2-3		3.17	4.5.19
102	Temperature Cycling	52-2-3		3.17	4.5.19
103	Fine Leak	52-2-3		3.17	4.5.19
104	Gross Leak	52-2-3		3.17	4.5.19
105	Electrical Test	52-2-3		3.17	4.5.19
106	Constant Acceleration	52-2-3		3.17	4.5.19
107	Fine Leak	52-2-3		3.17	4.5.19
108	Gross Leak	52-2-3		3.17	4.5.19
109	Electrical Test	52-2-3		3.17	4.5.19
Minor					
201	Evidence of poor workmanship		1.0%	3.21	Visual

NOTES:

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4.5 Test methods and procedures

4.5.1 Physical dimensions. The IC shall be tested for the requirements of Method 2016, of MIL-STD-883. Any IC failing to comply with all of the requirements of 3.4 shall be classified defective (Non-destructive test).

4.5.2 Bond strength. The IC shall be tested for the requirements of Method 2011, Test Condition D of MIL-STD-883 using Government approved test equipment. Any IC failing to comply with all of the requirements of 3.7 shall be classed defective (Destructive test).

4.5.3 Lead fatigue. The IC shall be tested for the requirements of Method 2004, Test Condition B2 of MIL-STD-883 using Government approved test equipment. Any IC failing to comply with all of the requirements of 3.10 shall be classed defective (Destructive test).

4.5.4 Die bond shear strength. The IC shall be tested for the requirements of Methid 2019, of MIL-STD-883 using Government approved test equipment. Any IC failing to comply with all of the requirements of 3.8 shall be classed defective (Destructive test).

4.5.5 Conditioning screentests. The IC shall be subjected to the required tests using Government approved test equipment. Any IC failing to comply with all of the requirements of 3.12 shall be classed defective. Testing is non-destructive and only IC's that pass all of the specified tests shall be returned to the lot.

4.5.6 Pre-Cap visual (DE-LID). The IC chip prior to capping shall be inspected for the requirements of Method 2001 Test Condition B of MIL-STD-883. Any “chip” failing to comply with all of the requirements of 3.9 shall be classed defective. (Destructive test)

4.5.7 Solderability. The IC shall be inspected for proper wetting as required in Method 2003 of MIL-STD-883. Any lead failing to comply with all of the requirements of 3.5 shall be classed defective (Destructive test).

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4.5.8 Resistance to solvents. The IC marking shall meet the requirements of Method 2016, of MIL-STD-883.Any package failing to comply with all of the requirements of 3.6 shall be classed defective (Destructive test).

4.5.9 Dynamic burn-in. All I.CI's shall be subjected to a dynamic burn-in test shall be performed at a temperature of one hundred twenty-five degrees centigrade (+125°C) for forty eight (48) hours with the IC connected as shown in figure 18. When the test is completed, the IC shall be cooled to twenty-five degrees (25°C) centigrade and the Electrical Tests delineated in Table II shall be performed.

4.5.10 Post burn-in electrical parameters. The test methods which follow are intended to convey the idea of what is to be tested and provide a simple bench method of doing it. It is recognized that automatic (computerized) tests will differ as to implementation of the test.

4.5 .10.1 Leakage current. The microcircuit shall be connected as shown in figure 9a. The supply voltage VDD shall be applied between terminals 16 and ground. A minimum of 50 milliseconds shall elapse before measuring the leakage current. After measuring the current with the microcircuit configured as shown in figure 9a, terminals 2 and 10 shall be switched from ground to VDD (See figure 9b). The leakage current with the microcircuit connected as shown in figure 9b shall be measured. Leakage currents shall be as specified in Table II.

4.5.10.2 Oscillator function current. The microcircuit shall be connected as shown in figure 4. VDD shall be applied to terminal 16 with terminal 4 connected to the delay network shown in figure 8a. After a minimum of 100 millisecond delay from application of VDD, terminal 4 shall be connected directly to VDD. The oscillator average current shall be measured over at least five (5) periods and shall be as specified in Table II. VDD rise time shall be 1.0 millisecond for all tests and sequential operation tests except long self destruct (SD) time test. A rise time of 500 milliseconds shall be used for the long SD time test.

4.5.10.3 Output period. With the microcircuit connected as shown in figure 4 and after the oscillator current has been measured, the Primary and Barrier oscillator periods shall be measured by connecting an approved frequency counter to terminals 3 and 11 respectively. The periods shall be as specified in Table II.

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4.5.10.4 Oscillator output voltage, terminals 3 and 11. With the microcircuit connected as shown in F16.4, the rise time, fall time and pulse duration of the pulse shall be measured on terminals 3 and 11 using a calibrated oscilloscope. Parameters shall meet the requirements listed in para. 3.14.4.

4.5.10.5 "0" output voltage: terminal 12. The microcircuit shall be connected as shown in figure 12. After application of VDD, 1040 pulses as defined in paragraph 3.3.6 shall be applied to terminal 2. The voltage level at terminal 12 shall be measured and shall be as specified in Table II.

4.5.11 Output voltage: terminals 13 and 15

4.5.11.1 "1" output voltage. The microcircuit shall be connected as shown in figure 10 and VDD applied. The initial low states at terminals 13 and 15 shall be-verified prior to the application of the appropriate number of input pulses, as defined in Note 2 page 37, to terminal 2. After application of 160 pulses, the voltage level at terminal 13 shall be measured. After 32, 768 pulses, the voltage level at terminal 15 shall be measured The voltages shall be as specified in Table II.

4.5.11.2 "0" output voltage. The microcircuit shall be connected as shown in figure 11. After application of VDD, any number of pulses less than 160 for terminal 13 or 32, 768 for terminal 15 shall be applied to terminal 2. The voltages levels at terminals 13 and 15 shall be measured and shalI be as specified in Table II.

4.5.12 Input switching voltage: terminals 2 and 10. The microcircuit shall be connected as shown in figure 14. A ramp generator with a convenient increasing voltage shall be applied to terminals 2 or 10. The voltage level at terminals 1 or 9, respectively, shall be monitored for a transition from a low to high level. When the transition occurs, the voltage level at terminals 2 or 10 shall be observed and measured as the threshold voltage. The threshold voltage shall be as specified in Table II.

4.5.13 Sequential operation

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4.5.13.1 Input requirements. The microcircuit shall be connected as shown in figure 5. The waveforms defined in figure 6 shall be applied to the appropriate input terminals. All input pulse magnitudes shall be equal to VDD except onterminal 4. The voltage levels applied to input terminal 4 during the sequence test must be the maximum and minimum LVD voltages specified in Table II for all values of supply voltages (VDD).

4.5.13.2 Input clock characteristics. All pulse counts shall be referenced to the zero start time. One pulse count is equal to one pulse of the oscillator (terminals 1-2) and shall conform to the characteristics defined in paragraph 3.3.5.

4.5.14 Input terminal number four (4) LVD

4.5.14.1 VDD and pin 4 delay rise time. Pin 4 shall be connected to a delay network as shown in figure 8a. VDD rise time shall be set for one millisecond for all tests and sequential operation tests except long SD time test. A VDD rise time of 500 milliseconds shall be used for the long SD time test.

4.5.14.2 VDD and pin 4 rise time. Pin 4 shall be connected to VDD directly as shown in figure 8b. VDD rise time shall be set for one millisecond for all-tests and sequential operation tests except long SD time test. A VDD rise time of 500 milliseconds shall be used for the long SD time test.

4.5.14.3 LVD function voltage: (terminal 15). The microcircuit shall be connected to VDD directly as shown in figure 13. A supply voltage (VDD) equal to 4.2 volts with a rise time of 1 millisecond shall be applied to terminal 16. A signal generator shall input 176 pulses as defined in 3.3.6 to terminal 2 to initialize the circuit. The voltage on terminals 4 and 16 shall be decreased at a convenient rate while the output voltage at terminal 15 is continuously monitored. When a voltage transition from low to high occurs at terminal 15, the magnitude of the supply voltage shall be measured as the LVD voltage and shall conform to the requirements of Table II.

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4.5.15 Input resistance. The microcircuit shall be connected as shown in figure 15. VDC shall be applied and the voltage from terminals 5, 6, and 7 to ground shall be measured. The voltages shall be as specified in Table II.

4.5.16 Output voltage, terminals 1, 3, 9 and 11

4.5.16.1 "1" output voltage. The output voltage on terminals 1, 3, 9, and 11 shall be measured with the microcircuit connected as shown in figure 16. Switch SW-1 shall be opened to measure the output voltage output on terminals 1 and 9. Switch SW-1, shall be closed and BW-2 be opened to measure the output on terminals 3 and 11. The voltages shall be as specified in Table II.

4.5,16.2 "0" output voltage. The output voltage on terminals 1, 3, 9, and 11 shall be measured by connecting the device as shown in Figure 14. Switch, SW-1, shall be opened and SW-2 closed to measure the voltage output on terminals 1 and 9. Switch SW-1, shall be closed and BW-2 shall be opened to measure that output on terminals 3 end 11. The voltages shall be as specified in Table II.

4.5.17 High termperature electrical parameters. The IC (First article tests shall be tested. at eighty degrees centigrade (+80°C) for the specified electrical requirements of Table IV. Any IC failing to comply with all of the specified requirements of Table IV shall be classified defective. Testing is non-destructive and only accepted IC's shall be returned to the lot. The IC (Lot acceptance tests) shall be performed at +60°C and shall meet the requirements of Table III.

4.5.18 Low temperature electrical parameters. The IC (First article tests) shall be tested at minus fifty five (-55°C) degrees centigrade for the specified electrical requirements of Table IV. The IC (Lot acceptance tests) shall be performed at -40°C and shall meet the requirements of Table III.

4.5.19 Group C environmental test. For requirements see paragraph 3.4. The integrated circuit shall be tested for the requirements of paragraph 4.4.2.6 on a continuing schedule of at least once every 90 days while the device is being produced. Testing shall be in accordance with MIL-STD-883. This testing is for informat ion only.

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4.5.20 Microcircuit mounting. The microcircuit device shall be mounted and soldered per MIL-S-45743 or MIL-S-46844 as applicable to the ADAM electronic board assembly.

4.5.21 ADAM wedge assembly. The electronic board assembly will be molded into a complete functionable ADAM wedge, using molding compound MC8196-01.

4.5.22 Pre-shock electrical tests. After molding, the microcircuit shall be electrically tested using the criteria set forth in MIL-P-48189.

4.5.23 Projectile loading. The test wedge assemblies shall be loaded into a 155MM Test Vehicle Projectile as defined by drawing 28010049 or 28010099. Modifications to the projectile base plate, ogive and to components internal to the ogive are allowed as long as those modifications maintain the same load conditions and packaging constraints on the wedge cargo.

4.5.24 Howitzer firing. The projectile shall be fired in a 155MM Howitzer utilizing an M109 or an M126 tube with a zone 7X equivalent firing. A zone 7X equivalent firing is defined as a firing wherein the breach pressure is 122% of the Recommend Mean Pressure (RMP) for a zone 7 firing, per Report No. AD826675L or APG Report No. 1375. A zone 7X propellant charge can be obtained by heating a zone 7 propellant charge to 145°F or by adding extra propellant as required by ambient temperature conditions.

4.5.25 Soft catch (Optional). Soft catch techniques which do not allow the projectile to tumble and apply less than a 5,000 G loading to the projectile may be used to recover the test vehicle after firing.

4.5.26 Flight ejection. When soft catch techniques are not used, the projectile fuze setting must be such that a cargo ejection height greater than 2000 feet is obtained. Ejection at lower altitudes may be considered grounds for no test.

4.5.27 Static election (Optional). Ejection and impact shock environments may be simulated on soft catch units. The simulated ejection will consist of firing the projectile expulsion charge, AR drawing 9216855, with the projectile mounted rigidly nose down is such a manner that the cargo is ejected into the air and allowed to fall back to the ground in a random orientation.

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4.5.28 Post shock testing. After the test vehicle has been subjected to the mechanical shock environments, wedges will be inspected for mechanical damage. The microcircuit will be tested in accordance with the Electrical Tests in MIL-P-48189(AR). Any failure attributable to the microcircuit is cause for rejection.

4.5.29 Process controls. The Government shall monitor the vendor's facility for compliance with MIL-M-38510. In addition, at random intervals, the Government Inspector shall monitor the contractor's quality and acceptability of metallization on the semiconductor dice using the scanning electron microscope. As a minimum, the following shall be monitored.

a. Metallization at all oxide steps.

b. General Metallization, defects such as peeling and voiding.

4.5.30 Data requirements. The contractor shall submit a quality inspection test, demonstration and evaluation report detail in accordance with Data Item Description DI-R-1724 (tailored\. A certificate of compliance shall be submitted to the technical agency listed in 6.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C. The Integrated Circuit shall be preserved and packaged in accordance with MIL-M-55565.

5.2 Marking. Marking of the carton shall comply with MIL-STD-1169.

5.3 Shipping. When integrated circuit packages of more than one lot are shipped together, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTE

6.1 Intented use. The integrated circuit covered by this specification is intended for use on as a low power timing base and function generator for use in scatterable mine assemblies.

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6.2 Ordering data. See Section 6.1 and 6.2 of MIL-A-48078.

6.3 Submission of inspection equipment designs for approval (see 6.2.3 of MIL-A-48078). Submit equipment designs as required to: Commander, U.S. Army-Armament Research and Development Command, ATTN: DRDAR-QAA-I, Dover, NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Distribution of certificate of compliance. distribution of the certificate of compliance for each IC lot shall be sent to:

Commander
ARRADCOM
ATTN: DRDAR-QAA-M
Dover, New Jersey 07801

6.6 Revision issue. The original of MIL-I-48331(AR) was never published.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1345-A116

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TABLE II

ELECTRICAL CHARACTERISTIC, ROOM AMBIENT, 25°C + 5°C

LOT ACCEPTANCE TESTS

CHARACTERISTIC	CONDITIONS		LIMITS
LEAKAGE CURRENT	VDD = 4.0 volts		1.0 u_a max.
OSCILLATOR FUNCTION CURRENT (SHALL BE THE AVERAGE VALUE MEASURED OVER AT LEAST FIVE (5) PERIODS	VDD = 4.0 volts		30 u_a max.
OSCILLATOR FUNCTION OUTPUT PERIOD	VDD = 4.0 volts		9 - 18 ms See Req't of Para. 3.2.7.3
"O" OUTPUT VOLTAGE TERMINAL 12	PIN 12 CONNECTED TO VDD THRU 10K OHMS	VDD = 3.0 volts	2.0 volts max.
"1" OUTPUT VOLTAGE TERMINALS 13, 15	PINS 13 & 15 CONNECTED TO GROUND THRU 1K OHM	VDD = 3.0 volts	1.5 volts min.
"0" OUTPUT VOLTAGE TERMINALS 13, 15	PINS 13 & 15 CONNECTED TO VDD THRU 1K OHM	VDD = 3.0 volts	1.5 volts max.
SEQUENTIAL OPERATION	VDD = 3.0 v or 0.5 v ABOVE LVD, WHICHEVER IS GREATER		
LVD FUNCTION VOLTAGE	TA = +25°C		1.7-2.9 volts
INPUT SWITCHING VOLTAGE TERMINALS 2 & 10	SEE PARA. 3.2.7.8		0.7-2.4 volts
INPUT RESISTANCE	TERMINALS 5, 6, & 7 CONNECTED TO VDD THROUGH 500K		2.0 volts min.

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TABLE II - CONT'D

"1" OUTPUT VOLTAGE TERMINALS 1,3,9,11	PIN CONNECTED TO GROUND THROUGH 50K OHMS	VDD = 3.0 volts SW_1 Open, SW_2 Open	1.5 volts min. Pins 3&11 1.5 v min.
		SW_1 Close, SW_2 Close	Pins 1&9 1.5 v min.
"0" OUTPUT VOLTAGE TERMINALS 1,3,9,11	PIN CONNECTED TO VDD THROUGH 50K OHMS	VDD = 3.0 volts SW_1 Open, SW_2 Open	1.5 volts max. Pins 1&9 1.5 v max.
		SW_1 Close, SW_2 Close	Pins 3&11 1.5 v max

NOTE:

ALL RESISTORS 1.0% METAL FILM, ALL CAPACITORS ± 10% TANTALUM, AND ALL VOLTAGES ± 0.05 VOLTS.

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TABLE III

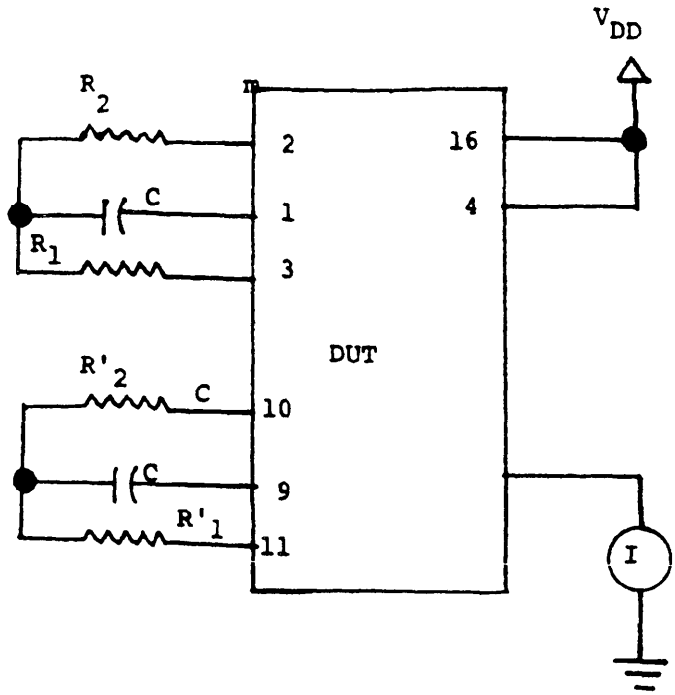
ELECTRICAL CHARACTERISTICS, HIGH & LOW TEMPERATURE, +60°C and -40°C

LOT ACCEPTANCE TESTS

CHARACTERISTIC	CONDITIONS	LIMITS
LEAKAGE CURRENT	TA = +60°C, VDD = 4.0v	16 u_a max.
OSCILLATOR FUNCTION CURRENT (SHALL BE THE AVERAGE VALUE MEASURED OVER AT LEAST FIVE (5) PERIODS	TA = -40°C, VDD = 4.0v TA = +60°C, VDD = 4.0v	50 u_a max. 50 u_a max.
OSCILLATOR FUNCTION OUTPUT PERIOD	F_1 F_2 TA = +25°C, VDD = 3.5v TA = +25°C, VDD = 4.0v TA = -40°C, VDD = 3.5v TA = -40°C, VDD = 4.0v TA = +60°C, VDD = 3.5v TA = +60°C, VDD = 4.0v	+5% F_1 & F_2 (25°C) -7% See Para. Req't (3.2.7.3)
"1" OUTPUT VOLTAGE TERMINALS 13, 15	TA = -40°C, VDD = 3.0v MEASURED AFTER APPLYING 4.0v THEN LOWER VDD to 3.0v	1.5 volts min.
"0" OUTPUT VOLTAGE TERMINALS 13, 15	TA = +60°C, VDD = 3.0v MEASURE AFTER APPLYING 4.0v, THEN LOWER VDD TO 3.0v AND BEFORE 176 PULSES HAVE BEEN APPLIED	1.5 volts max.
SEQUENTIAL OPERATION	TA = -40°C, VDD = 3.0v TA = -40°C, VDD = 4.5v TA = +60°C, VDD = 3.0v TA = +60°C, VDD = 4.5v	See Para. Req't (3.2.7.10)
LVD FUNCTION	TA = -40°C TA = +60°C	1.5 - 3.5 volts 1.5 - 3.5 volts
"0" OUTPUT VOLTAGE TERMINAL 12	TA = +60°C, VDD = 3.0v VDD = 3.0 VOLTS OR 0.1 VOLT ABOVE LVD WHICHEVER IS GREATER	2.0 volts max.

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OSCILLATOR FUNCTION



	NOMINAL VALUE	TOLERANCE	STYLE	TYPE	TCC
C	0.01 uF	± 1.0%	CERAMIC	NPO	± 25 PPM
R_1 & R'_1	470 K OHMS	± 1.0%	METAL FILM		± 50 PPM
R_2 & R'_2	2.0 MEGOHMS	± 5.0%	C.C.		

FIGURE 4

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SEQUENTIAL OPERATION

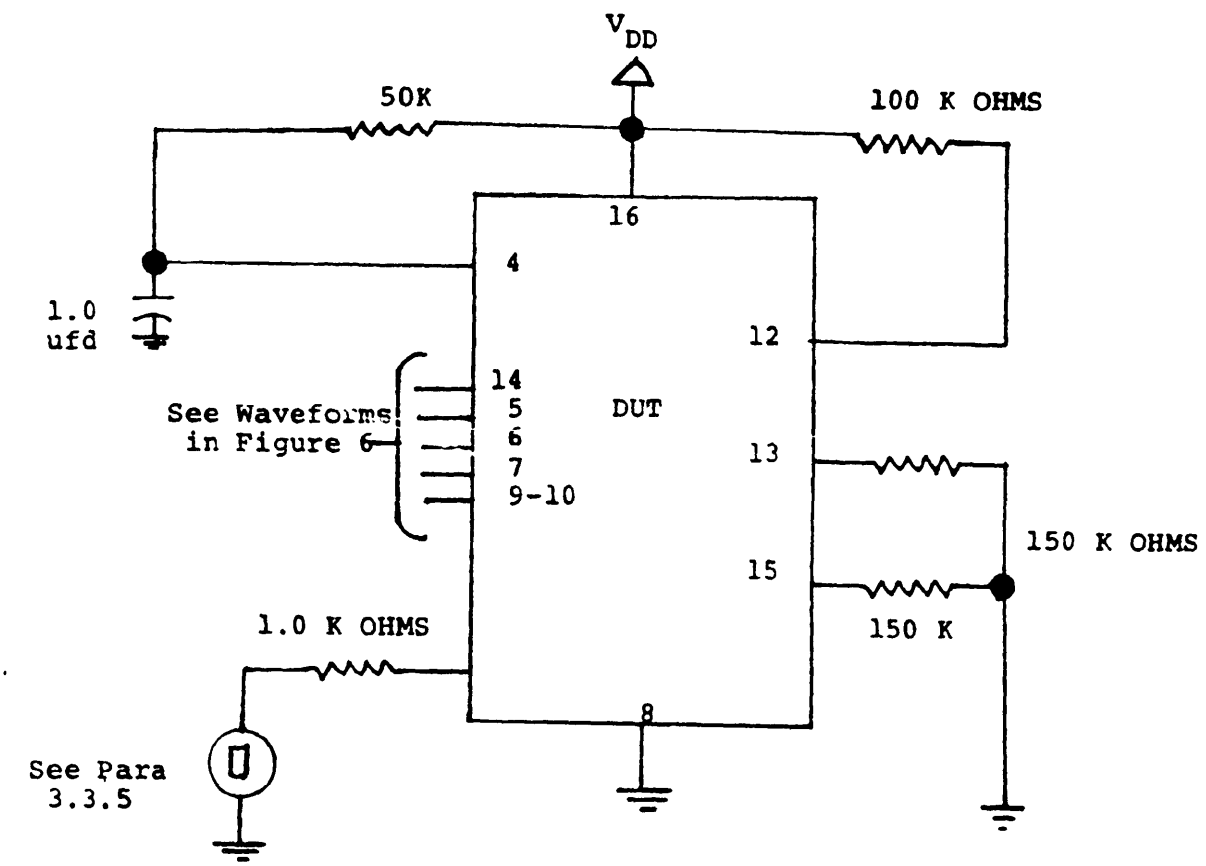


FIGURE 5

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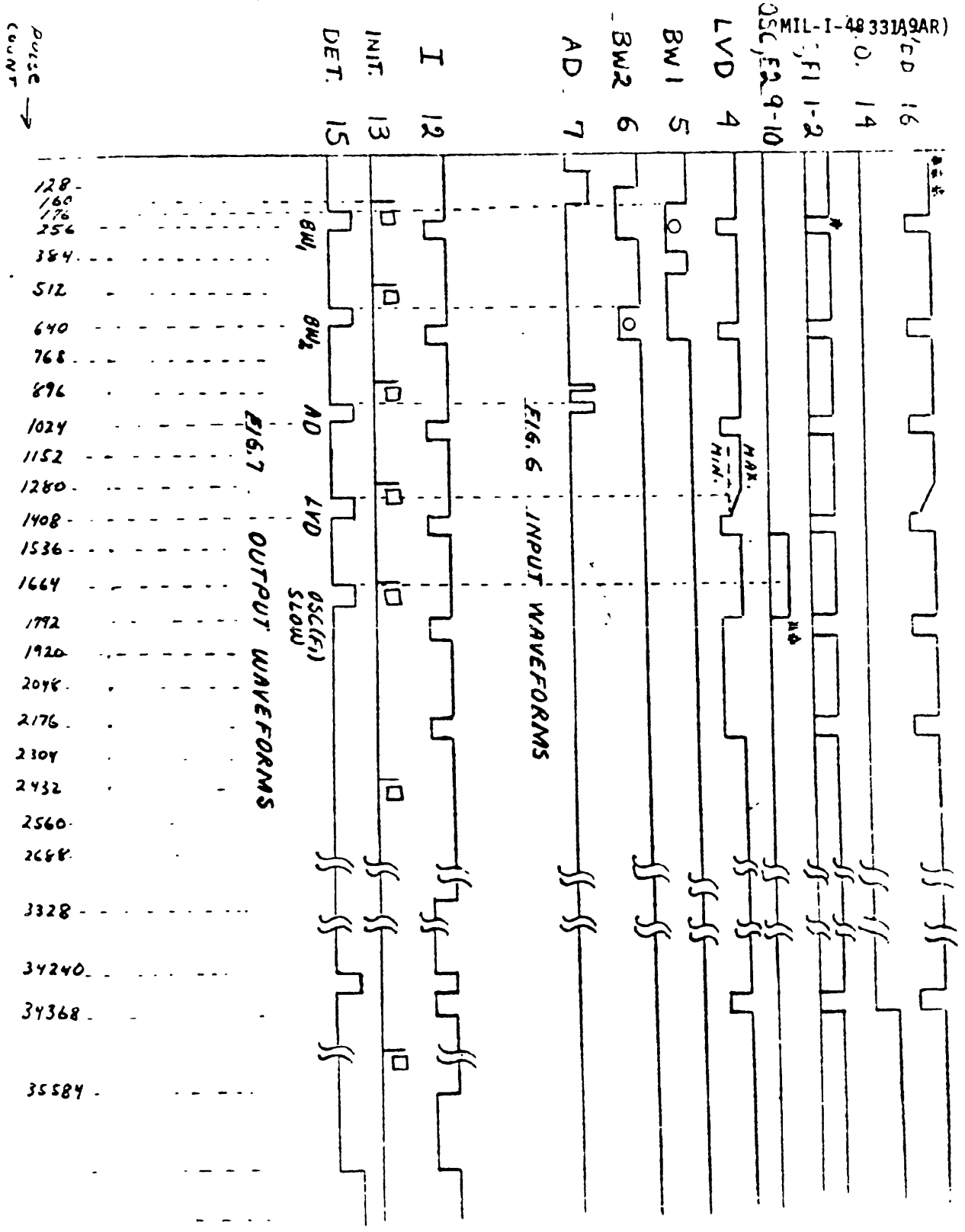


FIG. 6 INPUT WAVEFORMS

FIG. 7 OUTPUT WAVEFORMS

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NOTES on Fig. 6 and 7

1) All pulse magnitudes are equal to VDD, except on pin 4 as shown in Fig 6 (The input voltage levels applied to the input terminal 4 during the sequence test must be the max. and min. LVD voltages specified in Table II for all values of supply voltages (VDD)).

2) All pulse counts are referenced to the zero start time. One pulse counts is equal to one pulse of the oscillator (pins 1-2). Pulses shall be positive, with an amplitude equal to VDD and 524, 288.

Pulse Width = l microsecond min.
Rise Time = 1 microsecond max.
Decay Time = 1 microsecond max.

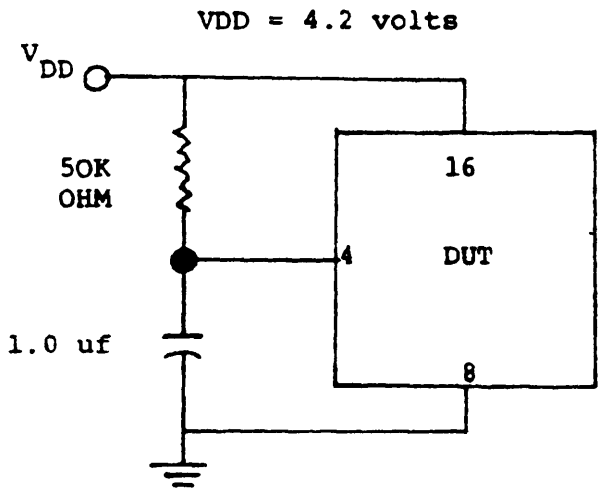
3) The I.C. shall function in accordance with this specification when pulsed with clock pulse recurrence rates between zero and 20 kilohertz. Function above 20 kilohertz is acceptable.

4) Symbol Terminology:

+ +	Pulse width is 2.0 micorseconds (3.0V VDD 4.5V)
⬭	Pulse width is 20 microseconds
▭	Pulse width is one input pulse
*	Repetitive pulse corresponding to input pulses
**	Pulses at 2X base input frequency
○	Removal of a high signal, not an applied low signal
***	VDD rise time shall be set to 1 millisecond and 500 millisecond in successive tests. Ground all IC pins when VDD=0 (Between tests)

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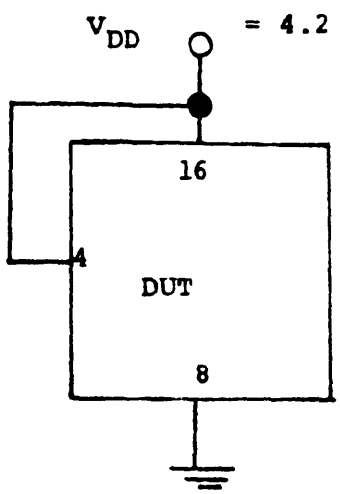
VDD AND PIN 4 DELAY RISE TIME



VDD RISE TIME SHALL BE SET TO 1.0 MILLISECONDS AND 500 MILLISECONDS IN SUCCESSIVE TESTS.

FIGURE 8a

VDD AND PIN 4 RISE TIME



VDD RISE TIME SHALL BE SET TO 1.0 MILLISECONDS AND 500 MILLISECONDS IN SUCCESSIVE TESTS.

FIGURE 8b

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LEAKAGE CURRENT

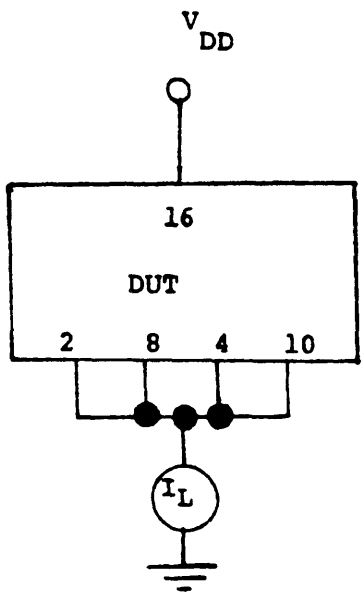


FIGURE 9A

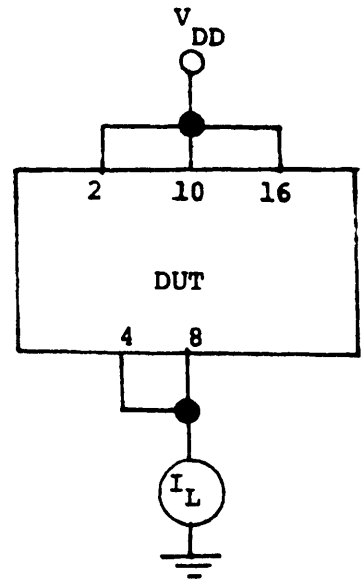


FIGURE 9B

OUTPUT VOLTAGE: TERMINALS 13 AND 15

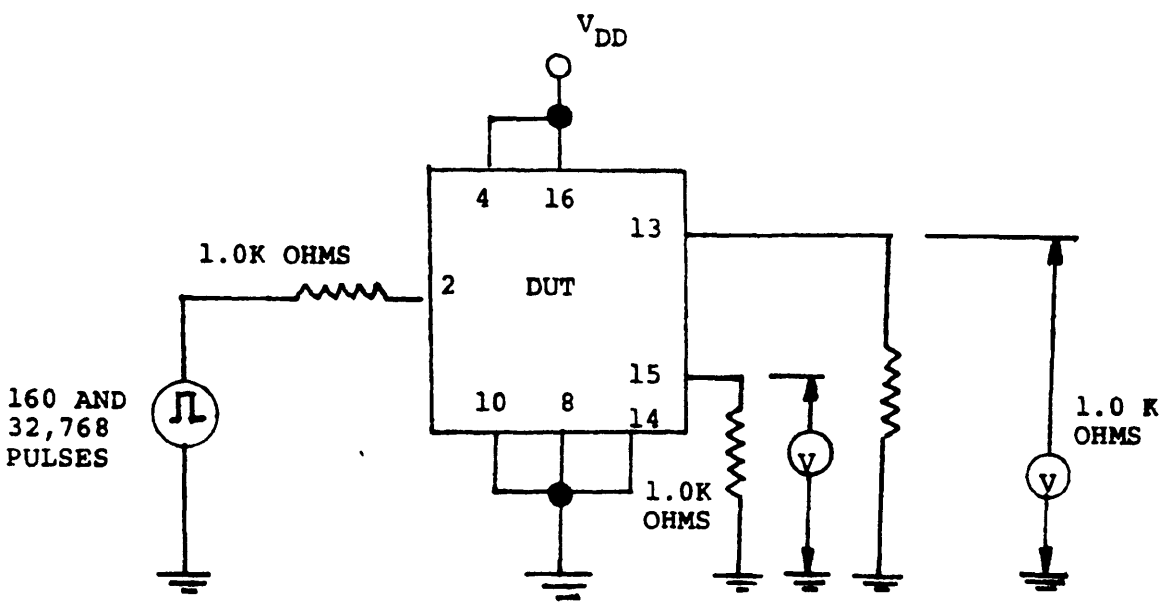


FIGURE 10

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"O" OUTPUT VOLTAGE ON TERMINALS 13 AND 15

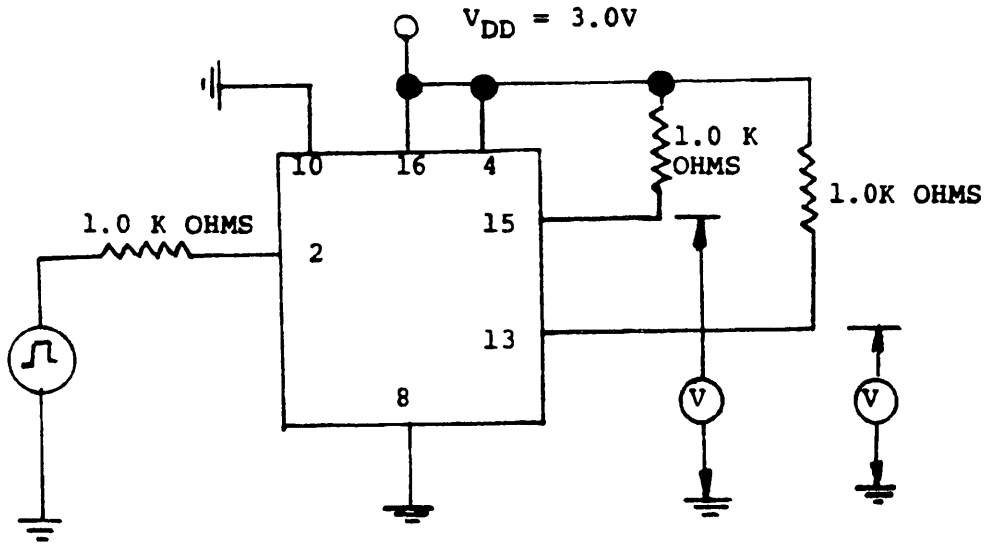


FIGURE 11

OUTPUT VOLTAGE: TERMINAL 12

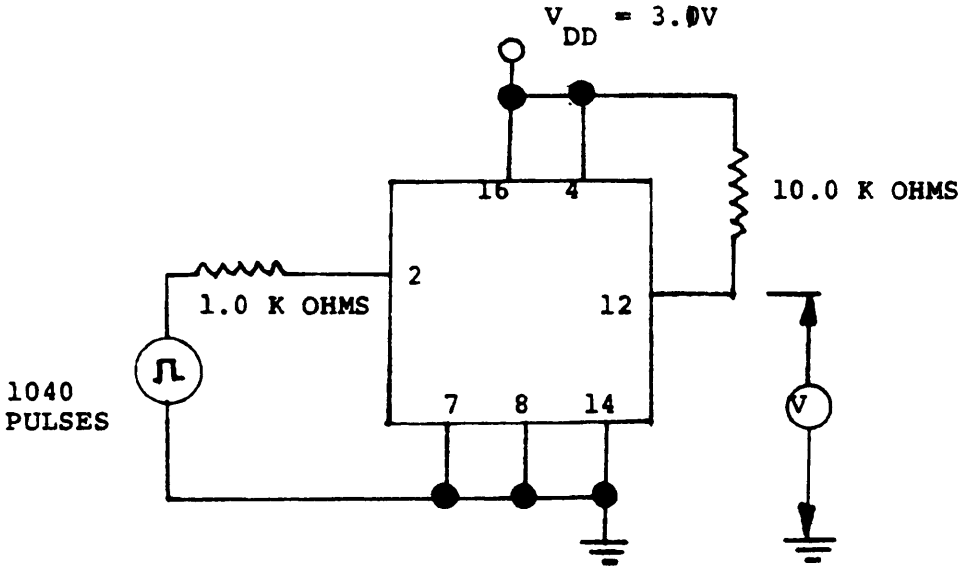


FIGURE 12

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LVD FUNCTION VOLTAGE TERMINAL 15

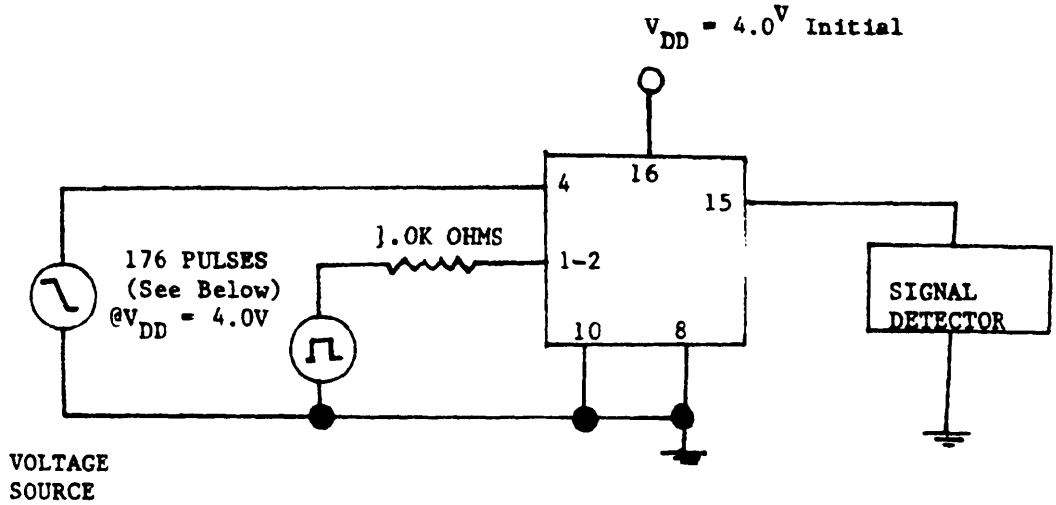
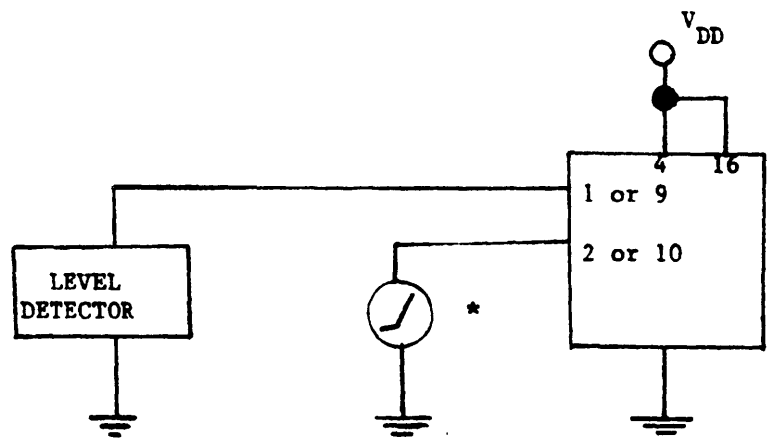


FIGURE 13

INPUT SWITCHING THRESHOLD VOLTAGE TERMINAL 2 AND 10



* Ramp Voltage Supply

FIGURE 14

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INPUT RESISTANCE

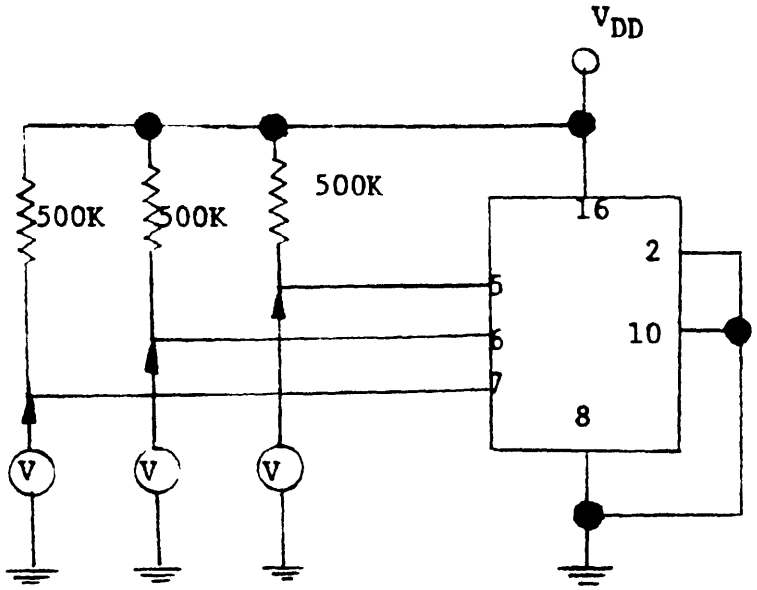


Figure 15

"1" OUTPUT VOLTAGE, TERMINALS 1, 3, 9, 11

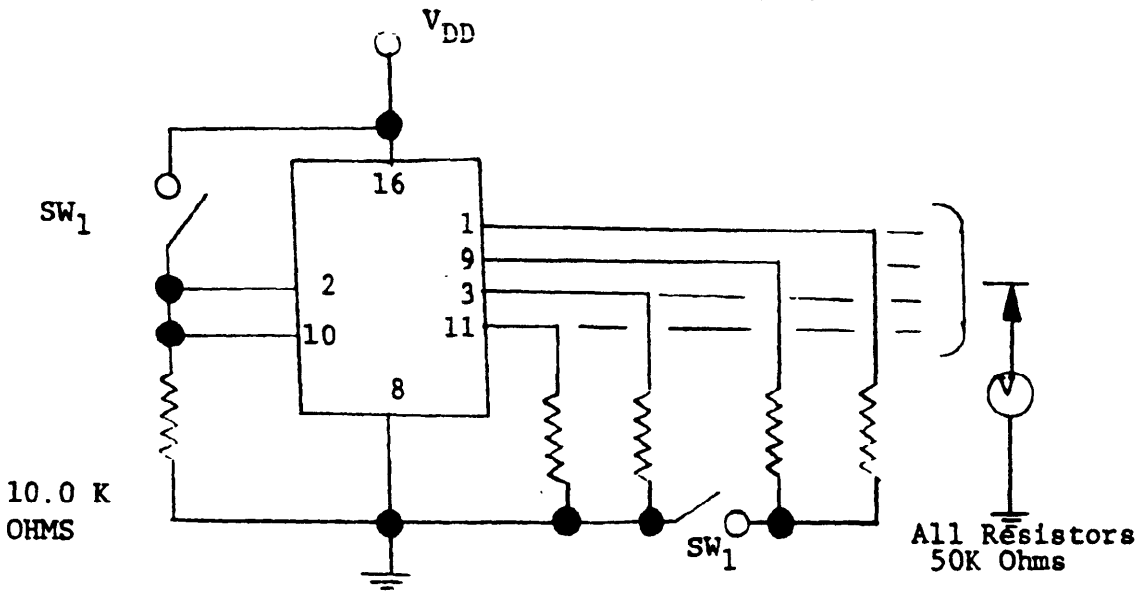
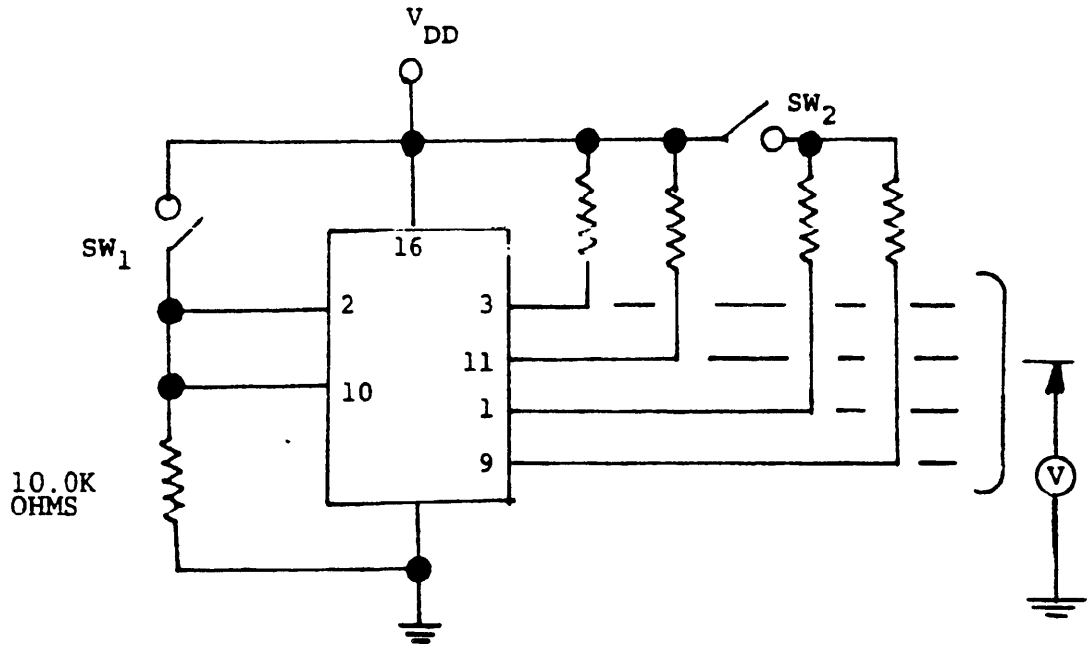


Figure 16

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"O" OUTPUT VOLTAGE TERMINALS 1, 3, 9, 11



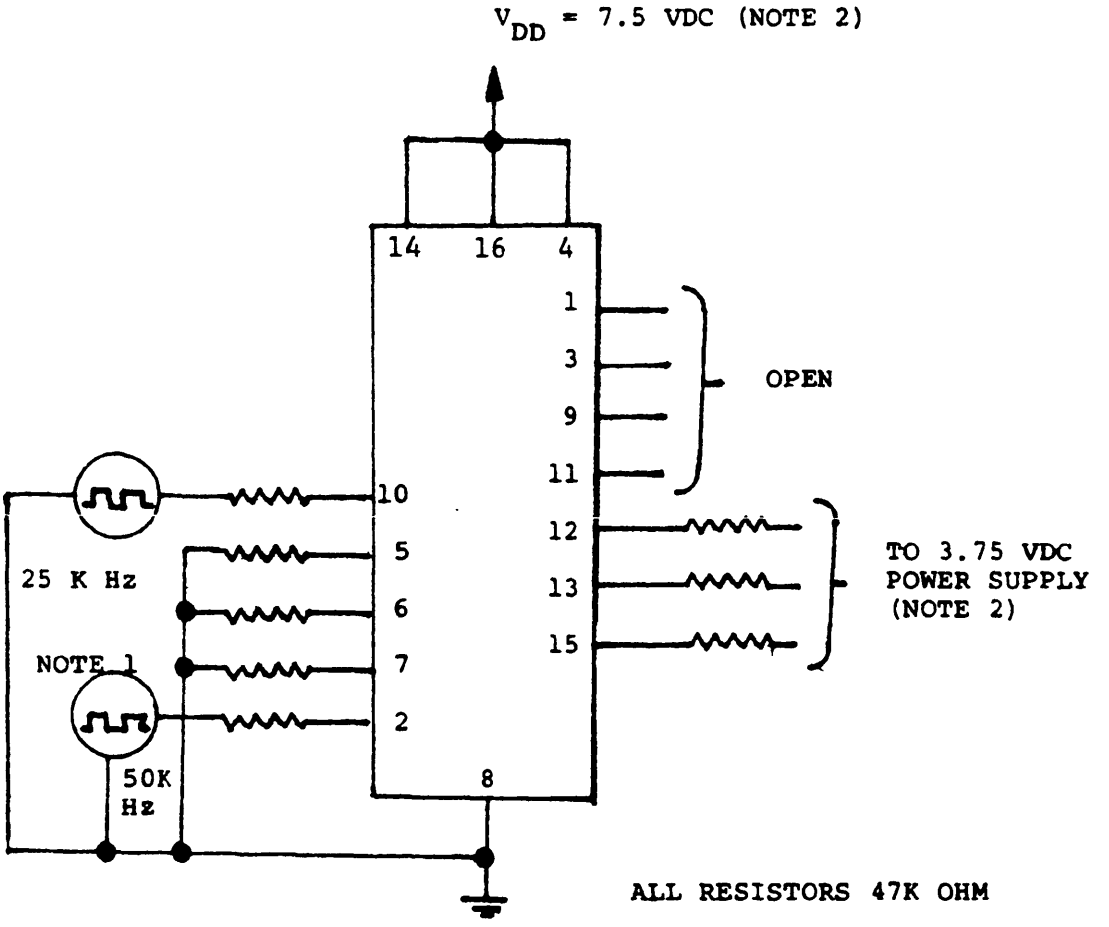
ALL RESISTORS 50K OHMS

FIGURE 17

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DYNAMIC BURN-IN TEST

A DYNAMIC BURN-IN TEST SHALL BE PERFORMED AT A TERMPERATURE OF 125°C WITH THE COMPONENT CONNECTED AS SHOWN IN FIGURE 18. WHEN THE TEST IS COMPLETED THE COMPONENT SHALL BE COOLED TO A TEMPERATURE OF 25°C. THEN THE TESTS IN TABLE II SHALL BE PERFORMED.



NOTES:

1. RACK FREQUENCY 30 - 70% DUTY CYCLE
   AMPLITUDE OF SQUARE WAVE: 0 AND 7.5V

2. BOTH POWER SUPPLIES SHALL BE TURNED OFF MOMENTARILY AT THE END OF EACH HOUR OF OPERATION TO ALLOW ALL LOGIC CIRCUITS TO RETURN TO THE RESET STATE.

FIGURE 18

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